

THE RISING SUN GREEN JOBS ARRIVE

With renewable energy being all the rage right now, solar literally produces the power to change the world. If you want a career in an industry that's only going to get bigger, solar may be for you

■ **MONIKA BHATI**

Blessed with one of the best quality and quantity of sun, India offers a huge potential to develop solar photovoltaic (PV) power both for domestic consumption and export. With several new companies from India and abroad starting their solar operations in India, and existing companies committing to expanding their capacity, solar PV power is bang in the middle of a take-off in India right now. The government too has been very supportive to boost the growth of solar industry. In January last year, the Ministry of New and Renewable Energy (MNRE) and Ministry of Power launched the Jawaharlar Nehru National Solar Mission (JNNSM) with an aim to set up an enabling environment for solar energy penetration in the country. The ambitious project aims to establish India as a global leader in solar energy, not just in terms of solar power generation but also in solar panels manufacturing and further development of this technology.

The JNNSM targets 22 GW of installed solar generation capacity by 2022, 100 GW by 2030 and 200 GW by 2050. It aims to achieve grid parity by 2022 and parity with thermal-based generation by 2030. It also aims to install 4-5 GW of solar manufacturing capacity in India by 2017. In the near term, the solar industry is looking to reach the total installed base of about 1 GW. Over the next two to three years, it may achieve 1-1.5 GW of annual installations year-on-year. Providing further impetus to the solar industry,

states like Gujarat and Rajasthan have announced their own solar policies.

Why solar?

Here's why you should think about making a career in this exciting field.

According to a recent KPMG report, solar power can meet 5-7 per cent of India's total power requirements by 2021-22, up from a negligible portion today. "Solar power can help our coun-

Job projection

As per the recent Confederation of Indian Industry (CII) study on manpower challenges submitted to MNRE, job projection for solar in 2022 is estimated at:

Solar PV on-grid	152,000
Solar PV off-grid	225,000
Solar thermal	270,000
Total	647,000 jobs

Source: HRD Strategies for Indian Renewable Energy Sector Report, October 2010, MNRE/CII

try move closer to the targeted 20-25 per cent reduction in carbon emission intensity of the total GDP by 2020, by contributing as much as one-tenth of this target, besides playing an increasingly important role in securing India's energy future," says Arvind Mahajan, head of energy and natural resources, KPMG.

Based on KPMG estimates, an average of 43,000 new jobs would be created annually in the period 2017-22 from the rooftop segment itself. Furthermore, around 42,000 jobs can potentially be created annually in the period 2017-22 from utility-scale solar power installations. The agriculture potential could create more than 38,000

jobs annually in the solar industry from 2017-18. KPMG further estimates that more than 600,000 jobs will be created during 2017-22 from rooftop, solar-powered agriculture pumpsets and utility-scale solar installations alone. Solar water heaters could also create more than 420,000 jobs in this period.

"Hence the solar industry would create close to a million jobs in the period 2017-22, carving a new industry segment like what IT has succeeded to create in the last decade in India," quips K. Subramanya, CEO, Tata BP Solar.

Solar industry, at the global level in general and national level in particular, is in a nascent stage hovering around 20GW per annum. "It is expected to grow manifold in years to come mainly due to two reasons: increasing concern for climate change necessitating switching to renewable energy from fossil fuels and also concern about the depleting oil reserves," adds Dr P. Jayakumar, CEO, Arbutus Consultants.

Exciting avenues

The solar industry offers exciting career opportunities in various disciplines of engineering like chemical engineering, mechanical engineering, electrical engineering, electronics and communications and civil engineering.

"If you check five years back, job opportunities were not even 10 per cent of what they are today. I personally see 30 per cent growth in this industry in the years to come," quips Hitesh C. Doshi, chairman, Waaree Group.

"With the launch of the NSM and